Software Engineering

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Problem

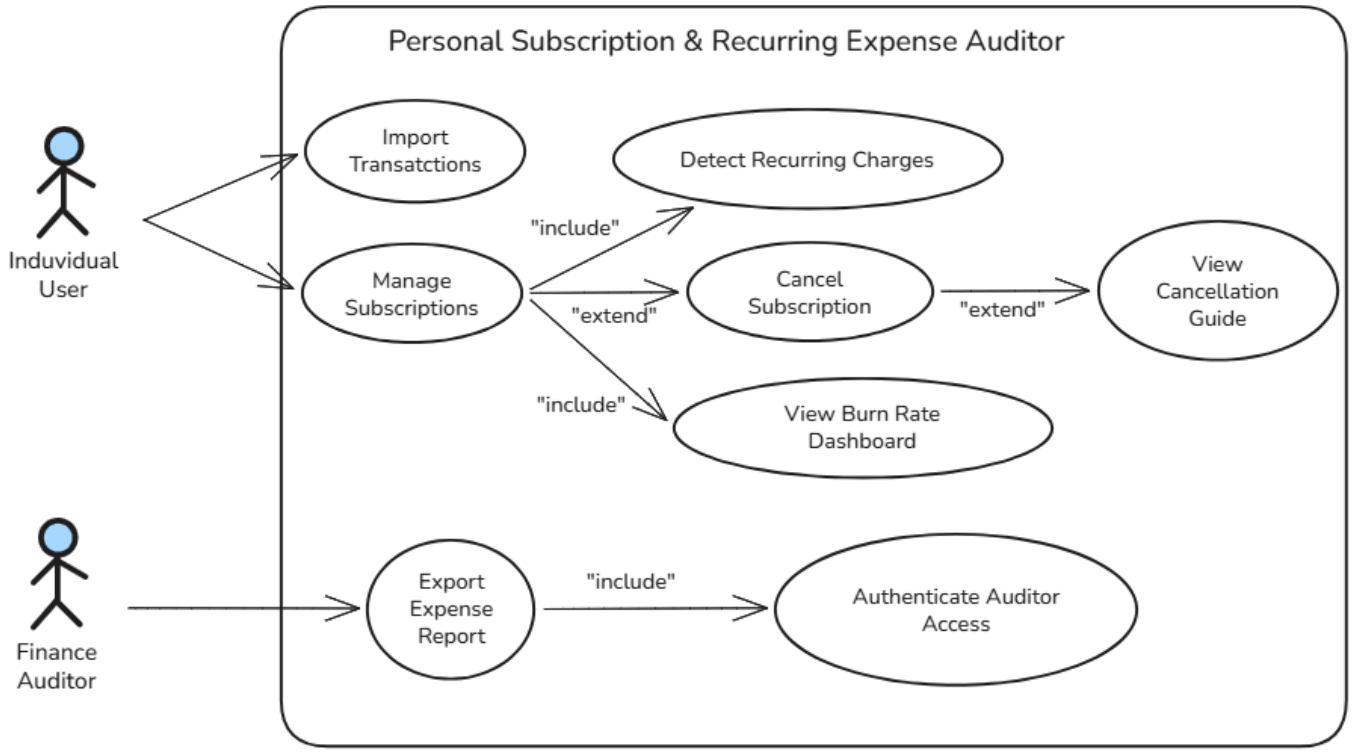
Personal Subscription & Recurring Expense Auditor

Function and Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall analyze transaction logs to identify recurring billing patterns and project total monthly subscription expenditure.	High	Pass: Recurring service identified and renewal date logged. Fail: One-time payment misclassified as subscription.	Forms the core detection engine that all other features (alerts, burn rate, cancellation) depend on.
FR-002	Functional	The system shall allow users to import transaction data via CSV/OFX file upload or connect a bank account through a secure aggregation API.	High	Pass: File parsed and transactions correctly mapped to standard fields (date, amount, merchant). Fail: Import fails silently or corrupts existing transaction history.	Enables the system to ingest the raw financial data needed for pattern analysis.
FR-003	Functional	The system shall notify users via email/push notification 5 days prior to an upcoming subscription renewal date.	Medium	Pass: Notification dispatched within 5 minutes of scheduled trigger. Fail: Notification fails to queue due to invalid user contact details.	Gives users enough lead time to decide whether to keep or cancel a subscription before being charged.

FR-004	Functional	The system shall calculate and display the user's total monthly and annual "burn rate" across all detected subscriptions, categorized by service type.	Medium	Pass: Dashboard total matches sum of active subscription charges. Fail: Cancelled or paused subscriptions still counted in burn rate.	Gives users clear visibility into their aggregate recurring spend for budgeting decisions.
FR-005	Functional	The system shall provide a one-click cancellation guide for a selected subscription, showing step-by-step instructions or a direct cancellation link for that merchant.	Low	Pass: Correct, up-to-date cancellation guide displayed for the selected merchant. Fail: Guide missing or links to an unrelated/incorrect merchant.	Reduces user friction in acting on the insights the app provides, closing the loop from detection to action.
NFR-001	Performance & Security	Financial transaction imports (CSV/OFX) must parse securely without storing raw banking credentials in cleartext.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Plaintext credentials detected in logs or database.	Protects sensitive banking data and maintains compliance with financial data-handling standards.
NFR-002	Security & Compliance	All stored transaction data and user financial profiles must be encrypted at rest and in transit using AES-256 and TLS 1.3 protocols.	High	Pass: Security audit and vulnerability scans report zero unencrypted sensitive payloads. Fail: Plaintext financial data detected in database logs.	Ensures regulatory compliance and protects users from data breaches involving sensitive financial information.

UML Use Case Diagram



Use Case Flow Specification

Core Use Case: Manage Subscriptions

Preconditions:

- User is authenticated and logged into the finance auditor app.
- User has at least one bank account or transaction file successfully linked/imported.
- Transaction data covers a minimum of one full billing cycle for pattern detection.

Postconditions (on success):

- Subscription database is updated with newly detected or modified recurring charges.
- Burn rate dashboard is recalculated to reflect current active subscriptions.
- Confirmation notification (if applicable) is sent to the user summarizing changes.

Main Success Scenario:

- User navigates to the "My Subscriptions" dashboard.
- System retrieves and displays all detected recurring charges, grouped by category, along with next renewal dates.
- User reviews the list and selects a subscription to inspect further.
- System «include» invokes Detect Recurring Charges to confirm the billing pattern is still active and consistent.
- User may choose to flag the subscription for cancellation.
- System «include» invokes View Burn Rate Dashboard to recalculate total monthly/annual spend excluding the flagged item.
- User confirms the action and clicks "Save Changes".
- System updates the subscription record and commits the database transaction.
- User receives a confirmation notification summarizing the updated subscription list and new burn rate.
- Use case ends successfully.

Alternate Flow – 5a. Misclassified Transaction Correction

- User identifies that a transaction has been incorrectly flagged as a recurring subscription (e.g., a one-time payment).
- System detects the user's manual override request on the classification.
- System removes the transaction from the recurring charges list and excludes it from burn rate calculations.
- System prompts the user to confirm whether similar future transactions from that merchant should also be excluded.
- System updates the merchant classification rules for that user's account.
- Use case ends successfully with corrected classification.